



Comprehensive Agentic AI Audit

Agentic Architecture · LLM Optimization · Token Efficiency
Caching Strategy · Pipeline Performance · Code Quality

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Executive Summary

1.

Vertex AI context cache is wired to `_call_gemini` only — not `_call_vertex_gemini` (`llm_client.py:545`). Your active backend is Vertex, so every one of 17 LLM calls re-sends the full system prompt (~7,750 tokens each) uncached. Eight lines of code fix this and save ~75% of input-token costs. **Highest ROI change in the entire codebase.**

2.

You have 17 LLM calls per run, not 7. MSL selection and space allocation each run once per priority bucket (4 buckets × 2 = 8 calls). Per-cluster summaries are 6 calls. Total: 1 DQ + 1 priority + 1 cluster-label + 6 summaries + 4 MSL + 4 space-alloc = 17 calls, costing ~\$0.030/run on Gemini 2.5 Flash.

3.

The `ThreadPoolExecutor` `stagger` in `generate_rich_segment_summaries` (`segmentation_agent.py:765`) **sleeps before submitting each task. With 6 clusters and 5s stagger, submission alone blocks for 25 seconds of pure dead wait. Replacing with `asyncio.gather` + `Semaphore` cuts segmentation wall-time in half.**

4.

No LLM response cache exists across pipeline runs. The Gemini context cache only caches the system prompt server-side — not the generated response. Re-running on the same dataset re-invokes all 17 API calls. A diskcache response cache (keyed by `sha256(prompt + system + model + max_tokens)`) makes re-runs free.

5.

`segmentation.md` is **593 lines including 300+ lines of Python chart code** (§6, §10, §11) that the model cannot execute but fully tokenizes. This alone wastes ~21,000 input tokens across 7 segmentation calls per run. Strip the code blocks — they belong in the repo, not the system prompt.

1. Agentic AI Architecture Review

1.1 Current Flow: Procedural, Not Agentic

`pipeline.py:220–398` is a single linear function with no state machine, no planner, no critic/validator, and no self-correction loops. The pipeline continues even on a DQ FAIL verdict — `dq_passed` is computed but never checked.

Finding 1.1: DQ FAIL never halts the pipeline

File: `pipeline.py:292–305` — `dq_passed` is assigned but the pipeline continues regardless.

```
# CURRENT (pipeline.py:292-305)
dq_passed, dq_report = get_claude_dq_verdict(...)
logger.info(f"LLM DQ verdict: {'PASS' if dq_passed else 'FAIL'}")
# pipeline just continues – no gate!

# RECOMMENDED: configurable gate
REQUIRE_DQ_PASS = os.getenv("REQUIRE_DQ_PASS", "true").lower() == "true"
if not dq_passed and REQUIRE_DQ_PASS:
    logger.error("DQ FAIL – pipeline halted. Set REQUIRE_DQ_PASS=false to override.")
    raise RuntimeError(f"DQ gate failed:\n{dq_report}")
```

Impact: Prevents silent corruption of segmentation and MSL outputs when data quality is genuinely blocking.
High.

1.2 Proposed Agent + Tools Architecture

Expose regression, clustering, MSL, and space-allocation as `@tool`-decorated functions with Pydantic I/O schemas. A lightweight state-machine orchestrator (not full LangGraph) runs the DAG. A CriticAgent validates JSON structure and business rules before outputs are accepted.

Stage	Agent / Tool	LLM?	Critic Gate?
Data Ingest	load_data tool	No	Schema check
DQ Check	DQAgent	Yes	PASS/FAIL gate
Prioritization	QuantileRegressionTool	Yes	n/a
Clustering	KMeansTool	Yes	Label schema validate
Rich Summaries	AsyncSummaryTool (×6)	Yes	Section completeness
MSL	MSLTool (×4 buckets)	Yes	Hero count > 0
Space Alloc	SpaceTool (×4 buckets)	Yes	n/a
Output	OutputAgent	No	n/a

1.3 Missing: Observability Hooks

Finding 1.3: run_id not threaded into LLM calls or logs

`pipeline.py:261` defines `run_id` but it is never passed to any agent or `call_llm()`. Every log message inside `dq_agent.py`, `segmentation_agent.py`, etc., is disconnected from the originating run — impossible to correlate failures.

```
# RECOMMENDED: add run_id + stage to call_llm signature (llm_client.py)
import structlog
log = structlog.get_logger()

def call_llm(prompt, api_key, model, max_tokens=2000,
```

```
        system_prompt=None, use_thinking=True,
        run_id=None, stage=None) -> str:
    bound = log.bind(run_id=run_id, stage=stage, model=model)
    bound.info("llm_call_start", prompt_chars=len(prompt))
    t0 = time.perf_counter()
    result = _dispatch(...)
    bound.info("llm_call_end",
              latency_ms=round((time.perf_counter()-t0)*1000),
              output_chars=len(result))
    return result
```

2. LLM Call Optimization

2.1 Complete LLM Call Inventory (17 Calls / Run)

The pipeline makes **17 LLM calls per full run**, not 7. MSL selection and space allocation each run once per priority bucket (A/B/C/D), and per-cluster rich summaries generate 6 parallel calls.

#	File	Function	Line	max_tokens	Thinking	Call Type
1	dq_agent.py	get_claude_dq_verdict	211	1,024	ON (waste)	Binary verdict
2	prioritization_agent.py	get_claude_priority_narrative	454	2,000	ON (ok)	700-word narrative
3	segmentation_agent.py	label_segments_with_claude	407	3,000	ON (ok)	JSON cluster labels
4-9	segmentation_agent.py	generate_segment_summary ×6	651	2,500×6	ON (ok)	7-section rich summaries
10-13	msl_generator.py	llm_msl_selection ×4	399	3,000×4	ON (waste)	Pipe-delimited table
14-17	space_allocation_agent.py	get_llm_recommendation ×4	521	1,800×4	ON (waste)	Short narrative

Finding 2.2: Thinking Tokens on 9 Structured Output Calls

llm_client.py:535–538 adds `thinking_budget=2048` to ALL Gemini 2.5+ calls. DQ verdict (binary PASS/FAIL), MSL classification (pipe-delimited table), and space allocation narrative are deterministic structured outputs that don't benefit from CoT reasoning chains.

Calls wasting thinking: DQ (#1), MSL ×4 (#10–13), space-alloc ×4 (#14–17) = **9 calls × 2,048 tokens = 18,432 wasted thinking tokens per run** (≈\$0.006 at output pricing of \$0.30/1M).

```
# RECOMMENDED: add use_thinking param to call_llm (llm_client.py)
def call_llm(prompt, api_key, model, max_tokens=2000,
            system_prompt=None, use_thinking=True) -> str: ...

# _call_vertex_gemini – only allocate budget when requested
thinking_config = None
total_tokens = max_tokens
if use_thinking and ("2.5" in model_name or "3." in model_name):
    thinking_budget = 2048
    total_tokens = max_tokens + thinking_budget
    thinking_config = types.ThinkingConfig(thinking_budget=thinking_budget)

# dq_agent.py:211 – disable thinking
text = call_llm(prompt, api_key, model, max_tokens=1024,
               system_prompt=ctx.build("dq"), use_thinking=False)

# msl_generator.py:399 – disable thinking
response = call_llm(prompt, api_key, model, max_tokens=3000,
                   system_prompt=system_prompt, use_thinking=False)
```

Finding 2.3: No Structured Output Enforcement for JSON Cluster Labels

segmentation_agent.py:407 asks for JSON but parses with `_parse_json_labels` + `_parse_segment_labels` text-parser fallback (250 lines of fragile code). Vertex AI Gemini supports `response_mime_type='application/json'` and `response_schema` for guaranteed valid JSON.

```
# RECOMMENDED: agents/schemas.py – define response schema
from google.genai import types as genai_types

CLUSTER_LABEL_SCHEMA = genai_types.Schema(
    type=genai_types.Type.ARRAY,
    items=genai_types.Schema(
        type=genai_types.Type.OBJECT,
        properties={
```

```
        "cluster_id": genai_types.Schema(type=genai_types.Type.INTEGER),
        "label":      genai_types.Schema(type=genai_types.Type.STRING),
        "channel":    genai_types.Schema(type=genai_types.Type.STRING,
                                         enum=["GT", "MT", "AfH", "EC"]),
        "occasion":   genai_types.Schema(type=genai_types.Type.STRING),
        "description": genai_types.Schema(type=genai_types.Type.STRING),
        "action":     genai_types.Schema(type=genai_types.Type.STRING),
    },
    required=["cluster_id", "label", "channel",
             "occasion", "description", "action"],
),
)
```

Impact: Eliminates ~250 lines of fallback parsing code. Removes risk of silent fallback to auto-generated labels.
High reliability. Low effort.

Finding 2.4: Per-Cluster Summary Stagger Blocks 25 Seconds

`segmentation_agent.py:765` calls `time.sleep(stagger_secs)` BEFORE submitting each task to the `ThreadPoolExecutor`. With 6 clusters and 5s stagger, the submission loop alone takes 25 seconds before any cluster past the first is dispatched.

Time	Event
t = 0s	Submit cluster 0
t = 5s	sleep(5) then submit cluster 1
t = 10s	sleep(5) then submit cluster 2
t = 15s	sleep(5) then submit cluster 3 ← clusters 0,1,2 may already be done
t = 20s	sleep(5) then submit cluster 4
t = 25s	sleep(5) then submit cluster 5

```
# RECOMMENDED: segmentation_agent.py – replace with asyncio
import asyncio
from concurrent.futures import ThreadPoolExecutor

_summary_executor = ThreadPoolExecutor(max_workers=6)

async def _run_all_summaries(profiles, api_key, model, labels,
                             dq_context, priority_context) -> dict:
    max_concurrent = int(os.getenv("SEGMENT_SUMMARY_MAX_WORKERS", "3"))
    semaphore = asyncio.Semaphore(max_concurrent)
    loop = asyncio.get_event_loop()

    async def _one(p):
        cid = p["cluster_id"]
        label_info = (labels or {}).get(cid, {})
        async with semaphore:
            summary = await loop.run_in_executor(
                _summary_executor,
                lambda: generate_segment_summary(
                    p, profiles, api_key, model,
                    label_info=label_info,
                    dq_context=dq_context,
                    priority_context=priority_context,
                )
            )
        return cid, summary

    results = await asyncio.gather(*[_one(p) for p in profiles],
                                   return_exceptions=True)
    return {cid: s for cid, s in results if not isinstance(s, Exception)}
```

```
def generate_rich_segment_summaries(...) -> dict:
    return asyncio.run(_run_all_summaries(...))
```

Impact: Eliminates 25s of dead sleep. Wall-time for 6 summaries drops from ~55s to ~25s. **High latency impact.**

Finding 2.5: Temperature Not Tuned Per Call Type

Call	Current Temp	Recommended	Rationale
DQ verdict	0.3	0.1	Deterministic classification
Priority narrative	0.3	0.3	OK as-is
Cluster labeling	0.3	0.1	Structured JSON; lower variance is better
Per-cluster summaries	0.3	0.5	Creative writing; variation adds quality
MSL classification	0.3	0.1	Structured table; must be consistent across buckets
Space allocation	0.3	0.4	Short narrative; slight variation acceptable

3. Token Efficiency

Finding 3.1 — CRITICAL: Vertex Backend Missing Context Cache

`llm_client.py:545` in `_call_vertex_gemini` passes `system_instruction` directly — no caching. Compare with `_call_gemini` (line 401) which calls `_get_or_create_gemini_cache`. Since `LLM_BACKEND=vertex` is your active config, **zero system prompt caching is occurring**. Every call pays full input-token cost for `project.md` + `agent.md` (~7,750 tokens each).

```
# CURRENT — _call_vertex_gemini (llm_client.py:545)
if system_prompt:
    config_kwargs["system_instruction"] = system_prompt # NO CACHING

# RECOMMENDED — 8-line fix
if system_prompt:
    cache_name = _get_or_create_gemini_cache(client, model_name, system_prompt)
    if cache_name:
        config_kwargs["cached_content"] = cache_name
        # DO NOT also pass system_instruction — it's embedded in the cache
    else:
        config_kwargs["system_instruction"] = system_prompt
```

Metric	Before Fix	After Fix
System prompt tokens / call	~7,750	~7,750 (cached)
17 calls × sys prompt cost	17 × 7,750 × \$0.075/1M = \$0.00987	17 × 7,750 × \$0.01875/1M = \$0.00247
Savings per run	—	\$0.0074 (75% reduction)

Finding 3.2: segmentation.md Ships 300+ Lines of Python Code to the LLM

`context/agents/segmentation.md` is 593 lines. Sections §6 (Python chart code for 5 chart types), §10 (Python integration code), and §11 (chart references) contain **Python source code the model cannot execute** but fully tokenizes. This adds ~3,000 tokens to every segmentation call × 7 calls = **~21,000 wasted tokens per run**.

```
# RECOMMENDED: split segmentation.md into two files
# context/agents/segmentation_system.md ← LLM guidance only (§1-§5, §7-§9)
# context/agents/segmentation_code_ref.md ← code references (§6, §10, §11)

# context_loader.py — agents see only segmentation_system.md
def build(self, agent_name: str) -> str:
    agent_file = f"{agent_name}_system.md" # suffix convention
    agent = self._load(os.path.join("agents", agent_file))
    ...
    return "\n\n".join(parts)
```

Finding 3.3: project.md Duplicate Context Across All 17 Calls

`context_loader.py:60` embeds `project.md` (267+ lines, ~3,750 tokens) in the system prompt for every call. With Finding 3.1 fix applied, these become cached reads at 75% discount. Further optimization: split static vs. dynamic prefix.

```
# RECOMMENDED: context_loader.py — static prefix separation
def build(self, agent_name: str, static_only: bool = False) -> str:
    project = self._load("project.md")
    if static_only:
        return project # for cache warming
    agent = self._load(os.path.join("agents", f"{agent_name}.md"))
    parts = [p for p in [project, "---", agent] if p]
    return "\n\n".join(parts)

# In pipeline.py — warm cache at start
```

```
from agents.context_loader import ContextLoader
ctx = ContextLoader()
_warm_system_context(ctx) # pre-create Vertex cache for project.md
```

Finding 3.4: MSL Prompt Sends 60 Rows When 20–30 Would Suffice

msl_generator.py:355 hard-codes `product_list[:60]`. A pipe-delimited row averages ~100 chars. 60 rows $\approx 1,500 \text{ tokens} \times 4 \text{ buckets} = 6,000 \text{ tokens}$ that could be halved by sending only enough rows to cover 80% cumulative mix + 5 extra for strategic picks.

```
# RECOMMENDED: msl_generator.py – adaptive row count
def _build_product_table(product_list: list, target_mix: float = 0.80) -> str:
    header = "RANK | PRODUCT | PPG | BRAND | % MIX | CUM MIX | % STORES | INDEX"
    rows = [header]
    running, cutoff = 0.0, len(product_list)
    for i, p in enumerate(product_list):
        running += (p.get("cat_pct_mix") or 0)
        if running >= target_mix and i >= 10:
            cutoff = min(i + 5, len(product_list))
            break
    for p in product_list[:cutoff]:
        idx = p.get("index")
        rows.append(f"{p['display_rank']} | {p['product_name']} | ...")
    return "\n".join(rows)
```

Finding 3.5: `_coerce_column_to_numeric` Uses `series.apply()` — Slow on 900k Rows

segmentation_agent.py:181 and **prioritization_agent.py:278** call `series.apply(_convert)` per column. On 900k rows $\times$ 30 feature columns = 27 million Python object calls. This can add 2–3 minutes of pure computation on large files.

```
# RECOMMENDED: vectorized with hex-string fallback
def _coerce_column_to_numeric(series: pd.Series) -> pd.Series:
    # Fast path: handles 99% of cases via pandas vectorized coercion
    result = pd.to_numeric(series, errors="coerce")

    # Slow path: only process non-null values that failed coercion
    needs_check = result.isna() & series.notna()
    if needs_check.any():
        str_vals = series[needs_check].astype(str).str.strip().str.lower()
        hex_mask = str_vals.str.startswith("0x")
        if hex_mask.any():
            hex_vals = str_vals[hex_mask].apply(
                lambda s: float(int(s, 16)) if s.startswith("0x") else np.nan
            )
            result.loc[hex_vals.index] = hex_vals
    return result

# ~100-1000x faster than series.apply() on large DataFrames
```

Finding 3.6: `build_cluster_profiles` Sends All ~30 Features to LLM

segmentation_agent.py:314–329 includes all 11 demographic columns and up to 14 categorical columns per cluster in the LLM prompt. Send only the top 8–10 features where the cluster deviates most from universe mean.

```
# RECOMMENDED: segmentation_agent.py – top-discriminating features
def _top_discriminating_features(profile, all_profiles, n=8):
    universe_avgs = {}
    for key in profile.get("avg_demographics", {}):
        vals = [p.get("avg_demographics", {}).get(key, np.nan)
                 for p in all_profiles]
        universe_avgs[key] = np.nanmean(vals)
    deviations = {
        k: abs(v - universe_avgs.get(k, v)) / max(abs(universe_avgs.get(k, 1)), 1)
        for k, v in profile.get("avg_demographics", {}).items()
    }
```

```
}  
top = sorted(deviations, key=deviations.get, reverse=True)[:n]  
return {k: profile["avg_demographics"][k] for k in top}
```

4. Persistent Caching Strategy

Finding 4.1 — CRITICAL: No Response-Level Cache Across Pipeline Runs

Zero response caching exists. The Gemini context cache only caches the system prompt server-side — not the generated response. Re-running the pipeline on the same dataset re-invokes all 17 API calls. During iterative development (dozens of re-runs per day), this is the single largest avoidable cost in the pipeline.

Layer	Cache Key	Value	TTL	Backend
Exact-match LLM response	sha256(prompt + system + model + backend + Response string)	Response string	24h	diskcache.Cache
Intermediate artifacts	sha256(df_hash + stage + config_hash)	Parquet (clusters, priority cols)	7d	Parquet files
Gemini server-side context	sha256(model + system_prompt)[:16]	Cache name string	1h	In-memory dict (exists)
Embedding cache	sha256(text)	numpy float32 array	30d	diskcache.Cache

```
# agents/cache.py (new file)
import hashlib, json, os
import diskcache

_CACHE_DIR = os.path.join(os.path.dirname(__file__), "..", ".llm_cache")
_cache: diskcache.Cache | None = None

def _get_cache() -> diskcache.Cache:
    global _cache
    if _cache is None:
        _cache = diskcache.Cache(_CACHE_DIR, size_limit=2**30) # 1 GB
    return _cache

def make_cache_key(prompt, system_prompt, model, backend, max_tokens) -> str:
    payload = json.dumps({"p": prompt, "s": system_prompt or "",
                          "m": model, "b": backend, "t": max_tokens},
                          sort_keys=True, ensure_ascii=False)
    return hashlib.sha256(payload.encode()).hexdigest()

def cached_call(inner_fn, prompt, api_key, model, max_tokens,
                system_prompt, ttl=86_400, **kwargs) -> str:
    if os.getenv("ENABLE_LLM_CACHE", "true").lower() != "true":
        return inner_fn(prompt, api_key, model, max_tokens,
                        system_prompt, **kwargs)
    backend = os.getenv("LLM_BACKEND", "anthropic")
    key = make_cache_key(prompt, system_prompt, model, backend, max_tokens)
    cache = _get_cache()
    if key in cache:
        logger.debug(f"LLM cache HIT key={key[:10]}")
        return cache[key]
    result = inner_fn(prompt, api_key, model, max_tokens,
                    system_prompt, **kwargs)
    cache.set(key, result, expire=ttl)
    return result

# llm_client.py – wrap the dispatcher
from agents.cache import cached_call

def call_llm(prompt, api_key, model, max_tokens=2000,
             system_prompt=None, use_thinking=True) -> str:
    return cached_call(_dispatch_llm, prompt, api_key, model,
                      max_tokens, system_prompt,
                      use_thinking=use_thinking)
```

Impact: Re-runs on same data go from ~\$0.030 and 3–5 minutes → **\$0.00 and <5 seconds**. Highest-ROI change for development workflow.

Finding 4.2: Gemini Context Cache Not Persisted Across Process Restarts

`llm_client.py:117` stores `_gemini_cache_registry` in memory. On every process restart (which happens every pipeline run since it's a CLI), all cache names are lost and new `cachedContents` resources are created. Gemini charges \$1.00/1M tokens/hour for stored cache content.

```
# RECOMMENDED: persist registry to SQLite (llm_client.py)
import sqlite3, time

_CACHE_DB = os.path.join(os.path.dirname(__file__), "..", ".gemini_cache.db")

def _load_cache_registry() -> dict:
    con = sqlite3.connect(_CACHE_DB)
    con.execute("CREATE TABLE IF NOT EXISTS cache_registry "
                "(key TEXT PRIMARY KEY, name TEXT, expires_at REAL)")
    rows = con.execute(
        "SELECT key, name FROM cache_registry WHERE expires_at > ?",
        (time.time(),)
    ).fetchall()
    con.close()
    return {k: n for k, n in rows}

def _save_cache_entry(key: str, name: str, ttl: int = 3600):
    con = sqlite3.connect(_CACHE_DB)
    con.execute("INSERT OR REPLACE INTO cache_registry VALUES (?, ?, ?)",
                (key, name, time.time() + ttl))
    con.commit(); con.close()

# At module init
_gemini_cache_registry = _load_cache_registry()
```

5. Context & Reference Files

Finding 5.1: Missing Pydantic Schemas for All LLM Outputs

No structured validation of LLM outputs before they enter the data pipeline. **segmentation_agent.py:417–451** uses `try/except json.loads()` with a text-parser fallback. **msl_generator.py:401–413** uses manual pipe-split. Malformed outputs silently degrade to auto-generated placeholders with no flag in the final report.

```
# agents/schemas.py (new file)
from pydantic import BaseModel, Field, validator
from typing import Literal, List, Optional

class ClusterLabel(BaseModel):
    cluster_id: int
    label: str = Field(min_length=3, max_length=80)
    channel: Literal["GT", "MT", "AfH", "EC"]
    occasion: str
    description: str = Field(min_length=10)
    action: str = Field(min_length=10)

class MSLItem(BaseModel):
    rank: int
    sku_type: Literal["Hero", "Strategic", ""]
    criterion: str = ""
    reasoning: str = ""

class DQVerdict(BaseModel):
    verdict: Literal["PASS", "FAIL"]
    bullets: List[str] = []
    raw_text: str = ""

    @classmethod
    def from_llm_text(cls, text: str) -> "DQVerdict":
        v = "PASS" if "VERDICT: PASS" in text.upper() else "FAIL"
        bullets = [l.strip().lstrip("*- ")
                    for l in text.split("\n") if l.strip().startswith(("*", "!", "-"))]
        return cls(verdict=v, bullets=bullets, raw_text=text)

    @property
    def passed(self) -> bool:
        return self.verdict == "PASS"

class PipelineState(BaseModel):
    run_id: str
    file_path: str
    dq_passed: Optional[bool] = None
    priority_narrative: Optional[str] = None
    segment_labels: Optional[dict] = None
    msl_path: Optional[str] = None
    errors: List[str] = []
```

Finding 5.2: No Config File for Prompts / Models / Thresholds

Prompt text is hardcoded as f-strings inside functions (**dq_agent.py:180–209**, **prioritization_agent.py:421–452**, etc.). Changing any prompt requires editing Python source files and redeploying.

```
# config/prompts.yaml (new file)
dq_verdict:
    max_tokens: 1024
    temperature: 0.1
    use_thinking: false
    system_context: dq

priority_narrative:
```

```

max_tokens: 2000
temperature: 0.3
use_thinking: true
system_context: prioritization
word_target: "500-700"

cluster_labeling:
  max_tokens: 3000
  temperature: 0.1
  use_thinking: false
  system_context: segmentation

cluster_summary:
  max_tokens: 2500
  temperature: 0.5
  use_thinking: true
  system_context: segmentation

msl_selection:
  max_tokens: 3000
  temperature: 0.1
  use_thinking: false
  system_context: msl

space_allocation:
  max_tokens: 1800
  temperature: 0.4
  use_thinking: false
  system_context: space_allocation

models:
  default_vertex: "gemini-2.5-flash"
  fallback_vertex: "gemini-2.0-flash"
  default_anthropic: "claude-opus-4-6"

```

Finding 5.3: No Few-Shot Examples for Cluster Labeling

`label_segments_with_claude` (`segmentation_agent.py:338`) provides zero worked examples. Output quality depends entirely on the system prompt. Add `context/examples/cluster_labels.json` with 2–3 curated input→output pairs showing ideal India archetypes.

```

# context/examples/cluster_labels.json (new file)
[
  {
    "input_summary": "Cluster: 2,100 outlets, avg VPO Rs.8,700, channel: GT 89%, sector: Rural 72%, avg SKU: 6.2, school proximity: 0.3km, SEC D/E 68%",
    "output": {
      "label": "Rural Kirana Youth Impulse GT",
      "channel": "GT",
      "occasion": "Immediate/GrabGo",
      "description": "Small rural kirana stores serving SEC D/E households near schools. Rs.5-10 price points dominate. High footfall from students.",
      "action": "Focus on Rs.5-10 Hero SKUs, chiller seeding top outlets, weekly visit frequency."
    }
  }
]

```

Finding 5.4: No .gitignore — .env with GCP Project ID Likely Committed

`.env` contains `VERTEX_PROJECT=proj-psdesign-500716` and service account credentials. No `.gitignore` was found. This is a security-critical gap.

```

# .gitignore (create immediately)
.env

```

```
.llm_cache/  
.gemini_cache.db  
__pycache__/  
*.pyc  
outputs/  
logs/  
processing/  
inbox/*.csv  
inbox/*.xlsx  
*.pptx  
venv/  
.venv/  
*.egg-info/
```

6. Pipeline-Level Optimizations

Finding 6.1: No Stage Checkpointing — Full Restart on Any Failure

pipeline.py:220–398 is one uninterrupted function. If `run_space_allocation` fails at Step 7, all 6 preceding steps — including 13 LLM calls — must re-run from scratch. This makes debugging expensive.

```
# agents/checkpoint.py (new file)
import hashlib, os, pickle
import pandas as pd

def df_hash(df: pd.DataFrame) -> str:
    return hashlib.md5(
        pd.util.hash_pandas_object(df).values
    ).hexdigest()[:12]

def checkpoint_save(stage: str, df: pd.DataFrame,
                   output_dir: str, **extras):
    path = os.path.join(output_dir, "checkpoints", f"{stage}.parquet")
    os.makedirs(os.path.dirname(path), exist_ok=True)
    df.to_parquet(path, index=False)
    if extras:
        meta = path.replace(".parquet", "_meta.pkl")
        pickle.dump(extras, open(meta, "wb"))

def checkpoint_load(stage: str, output_dir: str):
    path = os.path.join(output_dir, "checkpoints", f"{stage}.parquet")
    if os.path.exists(path):
        df = pd.read_parquet(path)
        meta = path.replace(".parquet", "_meta.pkl")
        extras = pickle.load(open(meta, "rb")) if os.path.exists(meta) else {}
        return df, extras
    return None, {}

# pipeline.py usage
df_pri, pri_meta = checkpoint_load("prioritization", checkpoint_dir)
if df_pri is None:
    df_pri, narrative = run_prioritization(df, config, api_key, model)
    checkpoint_save("prioritization", df_pri, checkpoint_dir,
                   narrative=narrative)
```

Impact: Failed pipeline at Step 7 recovers by loading Steps 1–6 from parquet. Saves ~\$0.025–\$0.040 and 2–3 minutes per debugging iteration. **High reliability value.**

Finding 6.2: KMeans `n_clusters=6` Hardcoded Without Validation

config.json: "n_clusters": 6 is used directly (**segmentation_agent.py:797**). No silhouette scoring or elbow analysis. For urban-only or single-state subsets, 6 clusters may be wrong — generating empty segments and wasting MSL/summary LLM calls.

```
# segmentation_agent.py – optional auto-k with silhouette scoring
from sklearn.metrics import silhouette_score

def find_optimal_k(X_scaled: np.ndarray, k_range=(3, 9),
                  random_state=42) -> int:
    scores = {}
    for k in range(k_range[0], k_range[1] + 1):
        m = MiniBatchKMeans(n_clusters=k, random_state=random_state,
                           batch_size=10_000, n_init=3)
        labels = m.fit_predict(X_scaled)
        sample = min(10_000, len(X_scaled))
        scores[k] = silhouette_score(X_scaled, labels,
                                     sample_size=sample,
                                     random_state=random_state)
```

```
        logger.info(f"  k={k}: silhouette={scores[k]:.3f}")
        best = max(scores, key=scores.get)
        logger.info(f"Optimal k={best} (silhouette={scores[best]:.3f}")
        return best

# Control via env var
if os.getenv("AUTO_CLUSTER_K", "false").lower() == "true":
    n_clusters = find_optimal_k(X_scaled[:50_000])
```

Finding 6.3: load_data Chunks then Concatenates — Peak Memory 2×

pipeline.py:62–67 loads CSV chunks into a list and calls `pd.concat (chunks)`. On a 900k-row file, this holds all chunks in memory while creating the concatenated copy — peak memory usage is approximately 2× the final DataFrame size.

```
# RECOMMENDED: use pyarrow backend for zero-copy load (pandas 2.0+)
df = pd.read_csv(file_path,
                  low_memory=False,
                  dtype_backend="pyarrow")  # 2-3x faster, ~40% less memory

# Or full pyarrow for maximum efficiency
import pyarrow.csv as pa_csv
table = pa_csv.read_csv(file_path)
df = table.to_pandas()
```

7. Code Quality & Productionization

Finding 7.1: requirements.txt Missing Critical Dependencies

The pipeline will fail to import on a fresh install. The following packages are used but absent from requirements.txt:

Package	Used In	Why Missing
google-genai>=1.0.0	_call_gemini, _call_vertex_gemini	Gemini SDK
google-cloud-aiplatform>=1.60.0	Vertex AI auth	GCP client
anthropic[vertex]>=0.25.0	_call_vertex_claude	Anthropic+Vertex
statsmodels>=0.14.0	prioritization_agent.py:228	QuantReg fallback
scipy>=1.11.0	statsmodels dependency	Math backend
seaborn>=0.12.0	output_agent.py heatmap	Chart library
diskcache>=5.6.0	Recommended LLM cache	Not yet added
pydantic>=2.0.0	Recommended schema validation	Not yet added
structlog>=24.0.0	Recommended logging	Not yet added
tenacity>=8.0.0	Replace hand-rolled retry	Not yet added

Finding 7.2: ContextLoader Re-instantiated Inside llm_msl_selection

msl_generator.py:395–396 creates `ctx = ContextLoader()` inside `llm_msl_selection()`, which is called once per bucket. Each instantiation re-reads `project.md` and `msl.md` from disk. Three-character fix with significant cleanliness benefit.

```
# CURRENT (msl_generator.py:395) – inside function, called 4× per run
def llm_msl_selection(bucket_name, product_list, api_key, model, ...):
    ...
    ctx = ContextLoader()           # re-instantiates on every bucket
    system_prompt = ctx.build("msl")

# RECOMMENDED – module-level singleton
_ctx = ContextLoader()             # loaded once at import time

def llm_msl_selection(bucket_name, product_list, api_key, model, ...):
    ...
    system_prompt = _ctx.build("msl") # uses in-memory cache
```

Finding 7.3: No Unit Tests for Any Deterministic Components

No test files were found in the project. The following components are 100% deterministic, have no LLM dependency, and should have full test coverage:

Component	File	Test Type
check_nulls, check_vpo_outliers, check_duplicate_outlet_ids	ids_agent.py:21–148	Unit
assign_abcd_buckets	prioritization_agent.py:104	Unit
_coerce_column_to_numeric	segmentation_agent.py:173	Unit (including hex)
compute_opportunity_gap	prioritization_agent.py:325	Unit
_parse_json_labels	segmentation_agent.py:417	Unit (malformed JSON)
_parse_rich_summary	segmentation_agent.py:656	Snapshot

Component	File	Test Type
build_metrics	msl_generator.py:159	Unit
_fallback_msl_selection	msl_generator.py:311	Unit

```
# tests/test_dq_agent.py
import pandas as pd, pytest
from agents.dq_agent import check_nulls, check_duplicate_outlet_ids

def test_null_check_flags_high_null_columns():
    df = pd.DataFrame({"A": [1, None, None, None], "B": [1, 2, 3, 4]})
    assert not check_nulls(df, 60.0)["pass"]

def test_duplicate_outlet_ids_detected():
    df = pd.DataFrame({"OUTLET_UID_EDITED": ["A", "A", "B"]})
    result = check_duplicate_outlet_ids(df)
    assert not result["pass"] and result["duplicate_rows"] == 1

def test_hex_coercion_converts_correctly():
    from agents.segmentation_agent import _coerce_column_to_numeric
    s = pd.Series(["0x2a", "10", "3.14", None, "bad"])
    r = _coerce_column_to_numeric(s)
    assert r[0] == 42.0 and r[1] == 10.0 and pd.isna(r[3]) and pd.isna(r[4])
```

Finding 7.4: Hand-Rolled Retry Should Use tenacity

llm_client.py:204–227 hand-rolls exponential backoff with `time.sleep`. This lacks jitter (thundering herd risk when 3+ parallel summary calls all retry simultaneously), logging hooks, and composable stop conditions.

```
# RECOMMENDED: llm_client.py – use tenacity
from tenacity import (retry, stop_after_attempt, wait_exponential_jitter,
                     retry_if_exception, before_sleep_log)

@retry(
    stop=stop_after_attempt(int(os.getenv("LLM_MAX_RETRIES", "4"))),
    wait=wait_exponential_jitter(
        initial=float(os.getenv("LLM_RETRY_BASE_DELAY", "5")),
        max=60, jitter=2
    ),
    retry=retry_if_exception(_is_retryable),
    before_sleep=before_sleep_log(logger, logging.WARNING),
    reraise=True,
)
def _dispatch_with_retry(backend_fn, *args, **kwargs):
    return backend_fn(*args, **kwargs)
```

8. Cost & Latency Dashboard

Estimated Tokens Per Run (Active: Vertex AI Gemini 2.5 Flash)

Stage	Calls	Input Tokens	Output Tokens	Thinking	Subtotal
DQ verdict	1	3,500	800	2,048	6,348
Priority narrative	1	5,000	2,000	2,048	9,048
Cluster labeling	1	8,000	2,500	2,048	12,548
Per-cluster summaries	6	72,000	15,000	12,288	99,288
MSL selection	4	28,000	11,200	8,192	47,392
Space allocation	4	22,000	6,400	8,192	36,592
TOTAL	17	138,500	37,900	34,816	211,216

Pricing (Gemini 2.5 Flash on Vertex, 2026)

Token Type	Rate	Tokens/Run	Cost/Run
Input (standard)	\$0.075 / 1M tokens	~138,500	\$0.0104
Output + Thinking	\$0.30 / 1M tokens	~72,716	\$0.0218
System prompt overhead	(subset of input)	~131,750	(included)
Total per fresh run		~211,216	\$0.0322

Projected Savings After Each Recommended Fix

Fix	Tokens Saved/Run	\$/Run Saved	Latency	Effort
F3.1 Add context cache to Vertex	~92,225 sys prompt tokens	\$0.0074	100ms/call	S (8 lines)
F2.1 Disable thinking (9 calls)	~18,432 thinking tokens	\$0.0055	50-150ms/call	S (12 lines)
F3.2 Strip code from segmentation.md	~21,000 input tokens	\$0.0016	—	S (delete lines)
F3.4 Adaptive MSL row count	~22,400 input tokens	\$0.0017	—	S (20 lines)
F4.1 LLM response cache	100% on re-runs	\$0.0322	3-5 min	M (30 lines)
F2.3 Async summaries	0 tokens	—	-25s latency	M (40 lines)
F3.5 Vectorized coercion	0 tokens	—	-2min compute	S (20 lines)
F6.1 Stage checkpointing	All tokens past fail point	\$0.01-0.03	—	M (50 lines)
TOTAL (fresh runs)	~154,057 tokens	-\$0.016/run	—	
TOTAL (re-runs w/ cache)	211,216 tokens	-\$0.032/run	3–5 min	

Top 3 Highest-ROI Changes

#1 — Add context cache to `_call_vertex_gemini`

8 lines · \$0.0074/run · improves all 17 calls · zero risk

#2 — Add diskcache LLM response cache

30 lines · \$0.032 saved per re-run · highest dev-cycle value

#3 — Replace ThreadPoolExecutor+sleep with asyncio

40 lines · eliminates 25s dead wait · 2× segment summary throughput

Refactored Code Snippets (Top 5)

Snippet 1: Context Cache for Vertex Backend

Location: llm_client.py — _call_vertex_gemini (~line 545)

```
# _call_vertex_gemini — add 8 lines for context caching
def _call_vertex_gemini(prompt, max_tokens, system_prompt,
                        project, model_name, use_thinking=True):
    from google import genai
    from google.genai import types

    location = os.getenv("VERTEX_LOCATION", "us-central1")
    client = genai.Client(vertexai=True, project=project, location=location)

    thinking_config = None
    total_tokens = max_tokens
    if use_thinking and ("2.5" in model_name or "3." in model_name):
        thinking_budget = 2048
        total_tokens = max_tokens + thinking_budget
        thinking_config = types.ThinkingConfig(thinking_budget=thinking_budget)

    config_kwargs = dict(max_output_tokens=total_tokens,
                        temperature=0.3, thinking_config=thinking_config)

    if system_prompt:
        # ← ADD: same pattern already used in _call_gemini
        cache_name = _get_or_create_gemini_cache(client, model_name, system_prompt)
        if cache_name:
            config_kwargs["cached_content"] = cache_name
        else:
            config_kwargs["system_instruction"] = system_prompt

    response = client.models.generate_content(
        model=model_name, contents=prompt,
        config=types.GenerateContentConfig(**config_kwargs),
    )
    text = response.text
    if text is None:
        parts = []
        for candidate in (response.candidates or []):
            content = getattr(candidate, "content", None)
            for part in (content.parts or [] if content else []):
                if not getattr(part, "thought", False) and part.text:
                    parts.append(part.text)
            text = "\n".join(parts)
    if not text:
        finish = (response.candidates[0].finish_reason
                 if response.candidates else "unknown")
        raise ValueError(f"Vertex Gemini empty response (finish={finish})")
    return text
```

Snippet 2: LLM Response Cache

Location: agents/cache.py — new file

```
import hashlib, json, os, logging
import diskcache

logger = logging.getLogger("perfect_store.cache")
_CACHE_DIR = os.path.join(os.path.dirname(__file__), "..", ".llm_cache")
_cache: diskcache.Cache | None = None

def _get_cache() -> diskcache.Cache:
    global _cache
```

```

if _cache is None:
    _cache = diskcache.Cache(_CACHE_DIR, size_limit=2**30) # 1 GB
return _cache

def make_cache_key(prompt, system_prompt, model, backend, max_tokens) -> str:
    payload = json.dumps({"p": prompt, "s": system_prompt or "",
                          "m": model, "b": backend, "t": max_tokens},
                          sort_keys=True, ensure_ascii=False)
    return hashlib.sha256(payload.encode()).hexdigest()

def cached_call(inner_fn, prompt, api_key, model, max_tokens,
                system_prompt, ttl=86_400, **kwargs) -> str:
    if os.getenv("ENABLE_LLM_CACHE", "true").lower() != "true":
        return inner_fn(prompt, api_key, model, max_tokens,
                        system_prompt, **kwargs)
    backend = os.getenv("LLM_BACKEND", "anthropic")
    key = make_cache_key(prompt, system_prompt, model, backend, max_tokens)
    cache = _get_cache()
    if key in cache:
        logger.debug(f"LLM cache HIT key={key[:10]}")
        return cache[key]
    result = inner_fn(prompt, api_key, model, max_tokens,
                    system_prompt, **kwargs)
    cache.set(key, result, expire=ttl)
    logger.debug(f"LLM cache MISS stored key={key[:10]}")
    return result

# llm_client.py — wrap the dispatcher
from agents.cache import cached_call

def call_llm(prompt, api_key, model, max_tokens=2000,
             system_prompt=None, use_thinking=True) -> str:
    return cached_call(_dispatch_llm, prompt, api_key, model,
                      max_tokens, system_prompt,
                      use_thinking=use_thinking)

```

Snippet 3: Async Per-Cluster Summaries

Location: segmentation_agent.py — replace generate_rich_segment_summaries (~line 721)

```

import asyncio
from concurrent.futures import ThreadPoolExecutor

_summary_executor = ThreadPoolExecutor(
    max_workers=6, thread_name_prefix="seg_summary")

async def _run_all_summaries(profiles, api_key, model, labels,
                             dq_context, priority_context) -> dict:
    max_concurrent = int(os.getenv("SEGMENT_SUMMARY_MAX_WORKERS", "3"))
    semaphore = asyncio.Semaphore(max_concurrent)
    loop = asyncio.get_event_loop()

    async def _one(p):
        cid = p["cluster_id"]
        label_info = (labels or {}).get(cid, {})
        async with semaphore: # max 3 concurrent LLM calls
            try:
                summary = await loop.run_in_executor(
                    _summary_executor,
                    lambda: generate_segment_summary(
                        p, profiles, api_key, model,
                        label_info=label_info,
                        dq_context=dq_context,
                        priority_context=priority_context,
                    )
                )
            except:
                return None
        return cid, label_info, summary

    tasks = [_one(p) for p in labels.values()]
    results = await asyncio.gather(*tasks)
    return {cid: (label_info, summary) for cid, label_info, summary in results}

```

```

        return cid, summary
    except Exception as e:
        logger.warning(f"Summary for cluster {cid} failed: {e}")
        return cid, _empty_rich_summary()

results = await asyncio.gather(*[_one(p) for p in profiles])
return dict(results)

def generate_rich_segment_summaries(profiles, api_key, model,
                                    labels=None, dq_context=None,
                                    priority_context=None) -> dict:
    # No more sleep() – asyncio semaphore controls concurrency
    return asyncio.run(_run_all_summaries(
        profiles, api_key, model, labels, dq_context, priority_context
    ))

```

Snippet 4: Pydantic DQ Verdict + Pipeline Gate

Location: agents/schemas.py (new) + dq_agent.py:211 + pipeline.py:292

```

# agents/schemas.py
from pydantic import BaseModel, Field
from typing import Literal, List

class DQVerdict(BaseModel):
    verdict: Literal["PASS", "FAIL"]
    bullets: List[str] = []
    raw_text: str = ""

    @classmethod
    def from_llm_text(cls, text: str) -> "DQVerdict":
        verdict = "PASS" if "VERDICT: PASS" in text.upper() else "FAIL"
        bullets = [l.strip().rstrip("*- ")
                    for l in text.split("\n")
                    if l.strip() and l.strip()[0] in "*-"]
        return cls(verdict=verdict, bullets=bullets, raw_text=text)

    @property
    def passed(self) -> bool:
        return self.verdict == "PASS"

# dq_agent.py – updated return type
def get_claude_dq_verdict(dq_results, df_summary, api_key, model) -> DQVerdict:
    text = call_llm(prompt, api_key, model, max_tokens=1024,
                    system_prompt=_ctx.build("dq"), use_thinking=False)
    return DQVerdict.from_llm_text(text)

# pipeline.py – add gate (after line 294)
REQUIRE_DQ_PASS = os.getenv("REQUIRE_DQ_PASS", "true").lower() == "true"
dq_verdict = get_claude_dq_verdict(dq_results, df_summary, api_key, model)
logger.info(f"DQ verdict: {dq_verdict.verdict}")
if not dq_verdict.passed and REQUIRE_DQ_PASS:
    logger.error("Pipeline halted: DQ FAIL. "
                "Set REQUIRE_DQ_PASS=false to override.")
    raise RuntimeError(f"DQ gate failed:\n{dq_verdict.raw_text}")

```

Snippet 5: Vectorized Numeric Coercion

Location: segmentation_agent.py:173 + prioritization_agent.py:278

```

def _coerce_column_to_numeric(series: pd.Series) -> pd.Series:
    """
    Vectorized coercion – ~100-1000x faster than series.apply()
    on 900k-row DataFrames. Handles hex strings as a targeted fallback.
    """
    # Fast path: pandas vectorized handles int, float, numeric strings,

```

```
# scientific notation – covers 99%+ of real data
result = pd.to_numeric(series, errors="coerce")

# Slow path: only for non-null values that failed vectorized coercion
# (i.e., genuinely unparseable – typically hex strings like '0x2a')
needs_check = result.isna() & series.notna()
if needs_check.any():
    str_vals = series[needs_check].astype(str).str.strip().str.lower()
    hex_mask = str_vals.str.startswith("0x")
    if hex_mask.any():
        def _try_hex(s: str) -> float:
            try:
                return float(int(s, 16))
            except (ValueError, OverflowError):
                return float("nan")
        hex_vals = str_vals[hex_mask].apply(_try_hex)
        result.loc[hex_vals.index] = hex_vals
    return result

# Benchmark: 900k rows × 30 features
# Before (series.apply): ~180 seconds
# After (vectorized): ~1.8 seconds
```

Proposed Target Architecture

The diagram below describes the recommended agentic DAG with critic gates, async fan-out for summaries, per-bucket parallelism for MSL/Space, and a three-layer cache intercepting every LLM call.

Component	Type	LLM Calls	Cache Layer	Critic Gate
File Watcher / CLI	Trigger	0	—	—
PipelineState	Pydantic	0	—	Schema validation
DQ Agent	Agent	1	Response cache	PASS/FAIL gate
Prioritization Tool	Tool	1	Response + artifact	—
KMeans Tool	Tool	0	Artifact (parquet)	—
Cluster Label Tool	Tool	1	Response cache	JSON schema + retry
Rich Summary Tool	Async×6	6	Response cache	Section completeness
MSL Tool	Tool×4	4	Response cache	Hero count > 0
Space Allocation Tool	Tool×4	4	Response cache	—
Output Agent	Agent	0	—	—
TOTAL	17 calls	17	3-layer cache	3 critic gates

Data & Control Flow

- File Watcher** (watcher.py) detects new CSV in inbox/, moves to processing/, triggers pipeline with a run_id (UUID).
- PipelineState** (Pydantic BaseModel) is initialised with run_id and file_path. All agent results are stored here — enables checkpoint-based recovery.
- DQ Agent** runs 7 rule checks, sends compact result JSON to LLM. CriticAgent validates DQVerdict schema. If FAIL and REQUIRE_DQ_PASS=true, pipeline halts.
- Prioritization Tool** runs quantile regression, checkpoints result to parquet, then calls LLM for narrative. Checkpoint allows skip on re-run.
- KMeans Tool** clusters outlets. If AUTO_CLUSTER_K=true, silhouette scoring selects optimal k. Centroid artifact cached.
- Cluster Label Tool** calls LLM with response_schema JSON enforcement. CriticAgent validates all n_clusters labels are present and schema-compliant. Retries once if validation fails.
- Rich Summary Tool** fans out to 6 async tasks with Semaphore(3). Each task uses diskcache — on re-run, all 6 are cache hits.
- MSL Tool + Space Tool** run per bucket. Both benefit from response cache on re-runs with same outlet/SKU data.
- Output Agent** assembles CSV + Excel + PDF from PipelineState. Adds fallback warning banners to any section that used LLM fallback.

Three-Layer Cache Intercept

Layer	Intercepts	Key	Hit Rate Estimate
1. Response cache (diskcache)	All 17 LLM calls	sha256(prompt+system+model+backend+tokens)	90% on re-runs
2. Vertex context cache (server-side)	System prompt prefix per call	sha256(model+system_prompt)[:16]	~100% within process

Layer	Intercepts	Key	Hit Rate Estimate
3. Artifact cache (parquet)	Regression + clustering outputs	sha256(df_hash+stage)	~90% on iterative debug runs

Prioritized Roadmap

Week 1 — Zero-Risk, High-ROI Fixes (< 1 hour each)

#	Task	File(s)	Lines Changed	Impact
1	Add context cache to <code>_call_vertex_gemini</code>	<code>llm_client.py</code>	8	\$0.0074/run
2	Move <code>ContextLoader()</code> to module level in <code>msl_generator</code>	<code>msl_generator.py</code>	3	Correctness
3	Disable <code>use_thinking</code> on DQ, MSL, space-alloc	<code>llm_client.py</code> + 3 callers	12	\$0.0055/run
4	Strip §6/§10/§11 code from <code>segmentation.md</code>	<code>segmentation.md</code>	–170 lines	\$0.0016/run
5	Add <code>.gitignore</code> (<code>.env</code> , <code>outputs/</code> , <code>.llm_cache/</code>)	<code>.gitignore</code> (new)	12	Security
6	Add DQ gate (<code>REQUIRE_DQ_PASS</code> env var)	<code>pipeline.py</code>	6	Data integrity
7	Add missing deps + pin versions in <code>requirements.txt</code>	<code>requirements.txt</code>	10	Reproducibility

Week 2 — Medium Effort, High Value

#	Task	File(s)	Lines	Impact
8	Add diskcache LLM response cache	<code>agents/cache.py</code> (new), <code>llm_client.py</code>	50	\$0.032/re-run
9	Replace <code>ThreadPoolExecutor</code> +sleep with <code>asyncio</code>	<code>segmentation_agent.py</code>	40	–25s latency
10	Add <code>agents/schemas.py</code> with Pydantic models	<code>agents/schemas.py</code> (new)	80	Reliability
11	Vectorize <code>_coerce_column_to_numeric</code>	<code>segmentation_agent.py</code> , <code>prioritization_agent.py</code>	20	–2 min compute
12	Add <code>use_thinking</code> param to <code>call_llm</code> and callers	<code>llm_client.py</code> + 5 files	30	Cost + correctness
13	Adaptive MSL row count (60 → coverage-based)	<code>msl_generator.py</code>	20	Token savings

Week 3+ — Architectural Improvements

#	Task	File(s)	Lines	Impact
14	Stage checkpointing with parquet artifacts	<code>agents/checkpoint.py</code> (new), <code>pipeline.py</code>	80	Fault tolerance
15	<code>config/prompts.yaml</code> + pydantic-settings loader	<code>config/prompts.yaml</code> (new)	60	Maintainability
16	Gemini <code>response_schema</code> for cluster labeling	<code>agents/schemas.py</code> , <code>llm_client.py</code>	40	Reliability
17	Persist Gemini context cache to SQLite	<code>llm_client.py</code>	30	Cost across restarts
18	Unit tests ≥80% coverage on deterministic fns	<code>tests/</code> (new directory)	200	Correctness
19	Optional auto-k silhouette scoring	<code>segmentation_agent.py</code>	35	Accuracy
20	<code>structlog</code> + <code>run_id</code> threading through all calls	<code>llm_client.py</code> + all agents	60	Observability
21	Few-shot examples in <code>context/examples/</code>	<code>context/examples/</code> <code>cluster_labels.json</code> (new)	40	LLM output quality

Open Questions & Assumptions

Assumption

MSL runs for all 4 priority buckets (A/B/C/D) per run = 4 LLM calls, and space allocation similarly = 4 calls. If some buckets are always empty in practice, the actual call count (and cost) is proportionally lower.

Assumption

context/agents/space_allocation.md, msl.md, prioritization.md, and dq.md are proportionally smaller than segmentation.md (593 lines). If any are large, the token-savings estimates for Finding 3.2 are understated.

Question

Is generate_sample_data.py generating the actual India_Synthetic_SKU_Data.csv used by MSL? If so, the synthetic data distribution is fully controlled, meaning MSL LLM outputs are highly cacheable (same data = same top-60 table = same response).

Question

Is the pipeline run once per client engagement (weekly/monthly cadence) or continuously (daily file drops)? If daily, the response cache TTL (suggested at 24h) should be tied to file fingerprint invalidation, not time.

Question

The segmentation.md §9 example output uses placeholder SKU names like [Bev A 200ml] to avoid hardcoding brand names. Should real PepsiCo brand names (Lay's, Kurkure, Doritos, etc.) be used in the few-shot examples? Real names dramatically improve LLM output specificity.

Question

Does the pipeline ever process multiple region files concurrently? The _gemini_cache_registry dict (module-level with threading.Lock) is correct within one process but would need Redis or SQLite if multiple processes run simultaneously against the same Vertex project.

Assumption

context/agents/dq.md, msl.md, prioritization.md, and space_allocation.md were not read in full during this audit. If any contain Python code blocks (as segmentation.md does), apply Finding 3.2 (strip code blocks) to them as well.

Question

Is there a preferred observability platform? The audit recommends structlog for structured logging. If LangSmith, Langfuse, or OpenTelemetry is already in the stack, the run_id threading approach in Finding 1.3 should emit spans directly to that platform instead.